

```
cd: no such file or directory: Desktop
[(base) murodjongafforov@Murodjons-MacBook-Pro /Users % cd--
zsh: command not found: cd--
[(base) murodjongafforov@Murodjons-MacBook-Pro /Users % cd --
[(base) murodjongafforov@Murodjons-MacBook-Pro ~ % cd Desktop
[(base) murodjongafforov@Murodjons-MacBook-Pro Desktop % mkdir example_project
[(base) murodjongafforov@Murodjons-MacBook-Pro Desktop % cd example_project
[(base) murodjongafforov@Murodjons-MacBook-Pro example_project % ls
[(base) murodjongafforov@Murodjons-MacBook-Pro example_project % pwd
/Users/murodjongafforov/Desktop/example_project
[(base) murodjongafforov@Murodjons-MacBook-Pro example_project % mkdir data
[(base) murodjongafforov@Murodjons-MacBook-Pro example_project % cd data
[(base) murodjongafforov@Murodjons-MacBook-Pro data % ls
[(base) murodjongafforov@Murodjons-MacBook-Pro data % pw
zsh: command not found: pw
[(base) murodjongafforov@Murodjons-MacBook-Pro data % pwd
/Users/murodjongafforov/Desktop/example_project/data
[(base) murodjongafforov@Murodjons-MacBook-Pro data % mkdir raw preprocessed engineered
[(base) murodjongafforov@Murodjons-MacBook-Pro data % ls
engineered      preprocessed    raw
[(base) murodjongafforov@Murodjons-MacBook-Pro data % touch README.md
[(base) murodjongafforov@Murodjons-MacBook-Pro data % cd ..
[(base) murodjongafforov@Murodjons-MacBook-Pro example_project % mkdir src
[(base) murodjongafforov@Murodjons-MacBook-Pro example_project % cd src
[(base) murodjongafforov@Murodjons-MacBook-Pro src % touch data.py preprocessed.py tuning.py models.py
[(base) murodjongafforov@Murodjons-MacBook-Pro src % ls
data.py          models.py        preprocessed.py  tuning.py
[(base) murodjongafforov@Murodjons-MacBook-Pro src % rm -r data.py
[(base) murodjongafforov@Murodjons-MacBook-Pro src % ls
models.py        preprocessed.py  tuning.py
[(base) murodjongafforov@Murodjons-MacBook-Pro src % cd ..
[(base) murodjongafforov@Murodjons-MacBook-Pro example_project % git mv models.py model.py
fatal: not a git repository (or any of the parent directories): .git
[(base) murodjongafforov@Murodjons-MacBook-Pro example_project % cd src
[(base) murodjongafforov@Murodjons-MacBook-Pro src % ls
models.py        preprocessed.py  tuning.py
[(base) murodjongafforov@Murodjons-MacBook-Pro src % git mv models.py model.py
fatal: not a git repository (or any of the parent directories): .git
[(base) murodjongafforov@Murodjons-MacBook-Pro src % .git mv models.py model.py
zsh: command not found: .git
[(base) murodjongafforov@Murodjons-MacBook-Pro src % mv models.py model.py
[(base) murodjongafforov@Murodjons-MacBook-Pro src % ls
model.py          preprocessed.py  tuning.py
[(base) murodjongafforov@Murodjons-MacBook-Pro src % █
```



```
[(base) muredjongafforov@Murodjons-MacBook-Pro src % git init
Initialized empty Git repository in /Users/muredjongafforov/Desktop/example_project/src/.git/
[(base) muredjongafforov@Murodjons-MacBook-Pro src % pwd
/Users/muredjongafforov/Desktop/example_project/src
[(base) muredjongafforov@Murodjons-MacBook-Pro src % cd ..
[(base) muredjongafforov@Murodjons-MacBook-Pro example_project % git init
Initialized empty Git repository in /Users/muredjongafforov/Desktop/example_project/.git/
[(base) muredjongafforov@Murodjons-MacBook-Pro example_project % ls
data    src
[(base) muredjongafforov@Murodjons-MacBook-Pro example_project % mkdir notebooks
[(base) muredjongafforov@Murodjons-MacBook-Pro example_project % cd notebooks
[(base) muredjongafforov@Murodjons-MacBook-Pro notebooks % pwd
/Users/muredjongafforov/Desktop/example_project/notebooks
[(base) muredjongafforov@Murodjons-MacBook-Pro notebooks % touch data_loading.ipynb
[(base) muredjongafforov@Murodjons-MacBook-Pro notebooks % git mv data_loading.ipynb experiment.ipynb
fatal: not under version control, source=notebooks/data_loading.ipynb, destination=notebooks/experiment.ipynb
[(base) muredjongafforov@Murodjons-MacBook-Pro notebooks % ls
data_loading.ipynb
[(base) muredjongafforov@Murodjons-MacBook-Pro notebooks % mv data_loading.ipynb experiment.ipynb
[(base) muredjongafforov@Murodjons-MacBook-Pro notebooks % ls
experiment.ipynb
[(base) muredjongafforov@Murodjons-MacBook-Pro notebooks % cd .
[(base) muredjongafforov@Murodjons-MacBook-Pro notebooks % cd ..
[(base) muredjongafforov@Murodjons-MacBook-Pro example_project % touch README.md
[(base) muredjongafforov@Murodjons-MacBook-Pro example_project % ls
README.md    data        notebooks    src
[(base) muredjongafforov@Murodjons-MacBook-Pro example_project % mkdir errors logs
[(base) muredjongafforov@Murodjons-MacBook-Pro example_project % ls
README.md    data        errors        logs        notebooks    src
[(base) muredjongafforov@Murodjons-MacBook-Pro example_project % touch requirements.txt
[(base) muredjongafforov@Murodjons-MacBook-Pro example_project % ls
README.md    data        errors        logs        notebooks
[(base) muredjongafforov@Murodjons-MacBook-Pro example_project % touch .gitignore .gitkeep
[(base) muredjongafforov@Murodjons-MacBook-Pro example_project % ls
README.md    data        errors        logs        notebooks
[(base) muredjongafforov@Murodjons-MacBook-Pro example_project % python version
python: can't open file '/Users/muredjongafforov/Desktop/example_project/version': [Errno 2] No such file or directory
[(base) muredjongafforov@Murodjons-MacBook-Pro example_project % python --version
Python 3.12.7
[(base) muredjongafforov@Murodjons-MacBook-Pro example_project % python -m venv example_project
[(base) muredjongafforov@Murodjons-MacBook-Pro example_project % pip list
Package                                Version
-----
aext-assistant                        4.1.0
```

uritemplate	4.2.0
urllib3	2.7.0
uvicorn	0.47.0
vega-datasets	0.9.0
w3lib	2.1.2
watchdog	4.0.1
wcwidth	0.2.5
webencodings	0.5.1
websocket-client	1.8.0
websockets	16.0
Werkzeug	3.0.3
whatthepatch	1.0.2
wheel	0.44.0
widgetsnbextension	3.6.6
wrapt	1.14.1
wurlitzer	3.0.2
xarray	2023.6.0
xgboost	3.2.0
xlwings	0.32.1
xyzservices	2022.9.0
yapf	0.40.2
yarl	1.11.0
zict	3.0.0
zipp	3.17.0
zope.interface	5.4.0
zstandard	0.23.0

```
[(base) muredjongafforov@Murodjons-MacBook-Pro example_project % pip freeze > requirements.txt]
```

```
[(base) muredjongafforov@Murodjons-MacBook-Pro example_project % pip list]
```

Package	Version
-----	-----
aext-assistant	4.1.0
aext-assistant-server	4.1.0
aext-core	4.1.0
aext-core-server	4.1.0
aext-panels	4.1.0
aext-panels-server	4.1.0
aext-project-filebrowser-server	4.1.0
aext-share-notebook	4.1.0
aext-share-notebook-server	4.1.0
aext-shared	4.1.0
aext-toolbox	4.1.0
aiobotocore	2.12.3
aiohappyeyeballs	2.4.0
aiohttp	3.10.5



```
[(base) muredjongafforov@Murodjons-MacBook-Pro example_project % source example_project/bin/activate
[(example_project) (base) muredjongafforov@Murodjons-MacBook-Pro example_project % diactivate
zsh: command not found: diactivate
[(example_project) (base) muredjongafforov@Murodjons-MacBook-Pro example_project % deactivate
(base) muredjongafforov@Murodjons-MacBook-Pro example_project % █
```

git clone https://github.com/A-Ahmed-I/ML-End-2-End



```
[(base) muredjongafforov@Murodjons-MacBook-Pro example_project % cd ..
[(base) muredjongafforov@Murodjons-MacBook-Pro Desktop % git clone https://github.com/A-Ahmed-I/ML-End-2-End
Cloning into 'ML-End-2-End'...
remote: Enumerating objects: 344, done.
remote: Counting objects: 100% (344/344), done.
remote: Compressing objects: 100% (235/235), done.
remote: Total 344 (delta 149), reused 268 (delta 89), pack-reused 0 (from 0)
Receiving objects: 100% (344/344), 9.87 MiB | 9.64 MiB/s, done.
Resolving deltas: 100% (149/149), done.
[(base) muredjongafforov@Murodjons-MacBook-Pro Desktop % ls
1-dars Modular Programming Report.pdf
1A SB audiolar @onlinetopik 4
1A WB @onlinetopik
2026-02-14 10.37.10.jpg
2026-07-11 14.41.53.jpg
2_dars_modular_programming
2_lesson_report_of_modular_programming.pdf
49464_51333_54617_45817_49892_50857_54620_44397_50612_50612_55064.pdf
Creative Formatting for Thesis by Gafforov Murodjon - Manus.html
F'rancais
Git_info.pdf
Hands_On_Machine_Learning_with_Scikit_Learn,_Keras,_and_TensorFlow.pdf
Heart Disease Prediction
Heart Disease Prediction.zip
Highlighted_Python_Data_Science_Handbook.pdf
ML Foundation
ML-End-2-End
Model Building
Model Improvement New Era of SML
Other Stuff
Prof. Kim Research Paper
Reportlar
Screenshot 2026-05-19 at 20.02.26.png
Screenshot 2026-06-22 at 16.12.08.png
Screenshot 2026-06-22 at 19.56.02.png
Screenshot 2026-06-23 at 17.51.06.png
Screenshot 2026-06-24 at 16.29.19.png
Screenshot 2026-06-24 at 16.31.13.png
Screenshot 2026-06-24 at 16.32.05.png
Screenshot 2026-06-24 at 18.16.51.png
Screenshot 2026-06-24 at 18.18.39.png
Screenshot 2026-06-26 at 17.32.04.png
Screenshot 2026-07-05 at 22.56.50.png
Screenshot 2026-07-06 at 01.30.29.png
```

Highlighted_Python_Data_Science_Handbook.pdf

ML Foundation

ML-End-2-End

Model Building

Model Improvement New Era of SML

Other Stuff

Prof. Kim Research Paper

Reportlar

Screenshot 2026-05-19 at 20.02.26.png

Screenshot 2026-06-22 at 16.12.08.png

Screenshot 2026-06-22 at 19.56.02.png

Screenshot 2026-06-23 at 17.51.06.png

Screenshot 2026-06-24 at 16.29.19.png

Screenshot 2026-06-24 at 16.31.13.png

Screenshot 2026-06-24 at 16.32.05.png

Screenshot 2026-06-24 at 18.16.51.png

Screenshot 2026-06-24 at 18.18.39.png

Screenshot 2026-06-26 at 17.32.04.png

Screenshot 2026-07-05 at 22.56.50.png

Screenshot 2026-07-06 at 01.30.29.png

Screenshot 2026-07-06 at 15.17.48.png

Screenshot 2026-07-07 at 17.56.13.png

Screenshot 2026-07-10 at 22.18.24.png

Screenshot 2026-07-11 at 14.08.08.png

Screenshot 2026-07-11 at 14.09.04.png

Screenshot 2026-07-11 at 15.40.27.png

The Books I Need to Learn

Thesis Paper Reference.pptx

UZAIRWAYS.jpg

[@Korea_Book] Lug'at 쓱 쓱 초급 TOPIK 1-2급 .pdf

[Korea_Book] Lug'at 쓱 쓱 초급 TOPIK 1-2급 .docx

dokumen_pub_hands_on_image_processing_with_python_1789343739_9781789343731 (1).pdf

epilepsy_prediction

example_project

for revision

git.docx

git_project1

heart_disease_presentation.pptx

original one.pptx

[(base) murodjongafforov@Murodjons-MacBook-Pro Desktop % cd ML-End-2-End

[(base) murodjongafforov@Murodjons-MacBook-Pro ML-End-2-End % ls

Data Science Salary	Health Care	LICENSE	Loan Status	README.md
---------------------	-------------	---------	-------------	-----------

[(base) murodjongafforov@Murodjons-MacBook-Pro ML-End-2-End % code .

[(base) murodjongafforov@Murodjons-MacBook-Pro ML-End-2-End % █

ML-End-2-End

healthcare_dataset.csv02_analytics.ipynb03_feature_engineering.ipynb04_feature_selection.ipynb

Health Care > notebooks > 04_feature_selection.ipynb > plot_features_importance(features_importance)

Generate + Code + Markdown Run All Clear All Outputs Outline Select Kernel

EXPLORER

ML-END-2-END

Health Care

data

processed

raw

healthcare_dataset.csv

deployment

model

modeling

notebooks

00_exploration.ipynb

01_clean data.ipynb

02_analytics.ipynb

03_feature_engineering.ipynb

04_feature_selection.ipynb

05_preprocessing.ipynb

06_test_models.ipynb

07_test_final_model.ipynb

reports

scripts

README.md

requirements.txt

setup.py

Loan Status

User Behavior

.gitignore

LICENSE

README.md

OUTLINE

TIMELINE

[1]

from scripts.feature_selection import encoder, train_random_forest, plot_features_importance
from scripts.configuration import EXPORT_PATH_2, EXPORT_PATH_3
from scripts.saving import save_data

Python

[2]

import pandas as pd

Generate + Code + Markdown Python

[3]

df = pd.read_csv(EXPORT_PATH_2)

Python

[4]

df.sample(5)

Python

...

	Age	Gender	Blood Type	Medical Condition	Date of Admission	Insurance Provider	Billing Amount	Room Number	Admission Type
30246	21	Male	O+	Diabetes	2020-09-02	UnitedHealthcare	13211.560722	255	Elective
15229	31	Female	B+	Asthma	2024-03-30	Blue Cross	23155.512273	132	Emergency
6206	38	Male	O-	Diabetes	2020-10-18	Cigna	24915.745887	288	Urgent
					2021-10-				

Cell 12 of 15

git log --> Bu kim, qachon va nima nom bilan commit qilganini ko'rsatadi. Asosan unda:

commit ID, muallif, sana va commit message bo'ladi.

A terminal window titled "ML-End-2-End — less ◀ git log — 170x44" displays the output of the "git log" command. The output shows a series of commits by "A7med Nabil" on "Wed Jun 4 01:17:17 2025 +0300". Each commit message is "Update README.md". The commit IDs are: 58bc5f2a54a207149a875a5a99295aaa0e503e93, a8b23b8e06b7f61eefcfec5f9b0521b24f0970a2, 29bf143ec3c9e84f6647ecadc8d4f0f69ed56011, 0edf24124faf67729c523be31f63dd628d565703, 634a4f3918e22e65d978f766195592acf4c90218, fcb40669344828bf6124b271973e2ab1cf91326a, 98c826646019b422a9c25a3e706597b6aaf9f4ed, and 9682481730ff48b7ab60f54887cdb318cb58c65b. The terminal window has a title bar with red, yellow, and green window control buttons.

```
[(base) murodjongafforov@Murodjons-MacBook-Pro ML-End-2-End % git log
commit 58bc5f2a54a207149a875a5a99295aaa0e503e93 (HEAD -> main, origin/main, origin/HEAD)
Author: A7med Nabil <120507686+A-A7med-i@users.noreply.github.com>
Date:   Wed Jun 4 01:17:17 2025 +0300

    Update README.md

commit a8b23b8e06b7f61eefcfec5f9b0521b24f0970a2
Author: A7med Nabil <120507686+A-A7med-i@users.noreply.github.com>
Date:   Wed Jun 4 01:15:47 2025 +0300

    Update README.md

commit 29bf143ec3c9e84f6647ecadc8d4f0f69ed56011
Author: A7med Nabil <120507686+A-A7med-i@users.noreply.github.com>
Date:   Wed Jun 4 01:14:04 2025 +0300

    Update README.md

commit 0edf24124faf67729c523be31f63dd628d565703
Author: A7med Nabil <120507686+A-A7med-i@users.noreply.github.com>
Date:   Wed Jun 4 01:13:41 2025 +0300

    Update README.md

commit 634a4f3918e22e65d978f766195592acf4c90218
Author: A7med Nabil <120507686+A-A7med-i@users.noreply.github.com>
Date:   Wed Jun 4 01:13:02 2025 +0300

    Update README.md

commit fcb40669344828bf6124b271973e2ab1cf91326a
Author: A7med Nabil <120507686+A-A7med-i@users.noreply.github.com>
Date:   Wed Jun 4 01:09:37 2025 +0300

    Update README.md

commit 98c826646019b422a9c25a3e706597b6aaf9f4ed
Author: A7med Nabil <120507686+A-A7med-i@users.noreply.github.com>
Date:   Wed Jun 4 01:07:24 2025 +0300

    Update README.md

commit 9682481730ff48b7ab60f54887cdb318cb58c65b
```

git diff --> Bu yerda biron faylni o'zgartirmaganiim uchun hech narsa ko'rinmadi aslida esa fayldagi kodlardan birontasi o'zgarsa uni ko'rsatib beruvchi command hisoblanadi.

```
(base) murodjongaafforov@Murodjons-MacBook-Pro ML-End-2-End % git diff
(base) murodjongaafforov@Murodjons-MacBook-Pro ML-End-2-End %
```

Bunda, (-) --> **minus** bu **o'chirilgan qator**, (+) -> **plyus** esa **qo'shilgan yoki o'zgargan qatordir**.

monkeytype --> bu tez yozishni mashq qilish va qo'lni qizdirib olish uchun:)

